

Test Plan

Smart Lab Equipment & Slot Reservation Portal

Document ID	TP-SLP-01
Version	1.0 (Mini-Project Part-1)
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Format	Organised in an IEEE 829-style test-plan structure as requested by the assignment (sections 1–5, §5.1 Security Validation); no formal standards-compliance is claimed
Inputs	SRS.md (SRS-SLP-01), Software_Architecture_and_Design.md, traceability/Requirements_Traceability.md

Execution status. The repository contains requirements and design documents only; the portal is not implemented yet. **No test case has been executed.** Every test case below has status *Not Executed* and an empty *Actual Result*. Results are to be filled in when an implementation exists.

1. Introduction

1.1 Purpose

This plan defines how the requirements of SRS-SLP-01 (FR-001...FR-010, NFR-001...NFR-003, SEC-REQ-01...SEC-REQ-07) will be verified, and contains the test cases that cover them.

1.2 Scope

Verification of the Smart Lab Equipment & Slot Reservation Portal at the system level through its HTTP API and web UI. Component-level (unit) testing is planned by the implementer once the technology stack is chosen and is not enumerated here.

1.3 Definitions

Terms and ID prefixes are those of the SRS §1.3. Additional: **Test clock** – a controllable time source used to create boundary situations (A-T2); **Stub** – a simulated Notification System.

1.4 References

SRS (SRS.md); Architecture & Design (Software_Architecture_and_Design.md); IEEE Std 829; assignment SE_Mini_Project_Deliverables_Part-1.pdf.

1.5 Overview

§2–§4 define what is tested; §5 the approach (including §5.1 Security Validation); §6–§11 environment, data, tools, roles, criteria and risks; §12 the test cases; §13 the traceability matrix.

2. Test Items

Item	Description	Source
TI-1	Smart Lab Equipment & Slot Reservation Portal – web UI and HTTP/JSON API (all components C-01... C-12)	SRS, Design §9
TI-2	Interface to the Notification System (verified against a stub)	SRS EIR-02
TI-3	Persistent store behaviour relevant to locking and audit records	SRS EIR-03

Version under test: *to be recorded at execution time* (no build exists yet).

3. Features to be Tested

Feature	Requirement	Use case	Test cases
View availability and reserve a slot (clauses a–e: availability, duration, horizon, overlap, token)	FR-001	UC-02, UC-03, UC-10	TC-01, TC-02, TC-03, TC-04, TC-17, TC-21, TC-24
Calibration check before confirmation	FR-002	UC-03, UC-09	TC-01, TC-05, TC-13
Cancellation and 5-second slot release	FR-003	UC-04	TC-06, TC-07, TC-23
Return recording and late-return flag	FR-004	UC-06, UC-12	TC-08, TC-09, TC-22
Equipment inventory add / update / remove	FR-005	UC-08	TC-10, TC-21, TC-22, TC-25
Login and role assignment	FR-006	UC-01	TC-11, TC-19, TC-20
Own reservation history	FR-007	UC-05	TC-12
Calibration status update	FR-008	UC-07	TC-13, TC-22
Confirmation display and notification	FR-009	UC-11	TC-14
Student self-overlap rejection	FR-010	UC-03	TC-15
Reservation latency ≤ 200 ms	NFR-001	UC-03	TC-16
RBAC and 30-minute session expiry	NFR-002	UC-06/07/08	TC-18, TC-19
No double booking under concurrency	NFR-003	UC-03	TC-17
Security requirements	SEC-REQ-01... 07	–	see §5.1

4. Features Not to be Tested

Item	Reason
User registration, password reset	Out of scope of the SRS (A-11).
Actual email / push delivery by the real Notification System	External system; only the request made by the portal is verified (stub, A-12).
Browser/device compatibility, visual layout, accessibility	No requirement specifies them (SRS §3.3 EIR-01).
Capacity beyond the A-03 test load, availability %, long-term endurance	No such requirement exists; none are invented.
Transport encryption (HTTPS) and password-hash algorithm	Deployment / implementation choices (A-10); to be reviewed as configuration, not by a functional test.
Penalties other than the late-return flag	Not defined in Lab 1.
Technician-initiated cancellation	Out of scope (A-09).

5. Approach

Testing is black-box and requirement-driven. Each test case traces to at least one SRS requirement. Test design techniques: equivalence partitioning (valid / invalid roles, calibration states), boundary-value analysis (120 / 121 min, 7 days, 60 / 59 min before start, 29 / 31 min idle, return at slot end), negative testing and fault injection (notification stub failure), concurrency testing, and security testing as in §5.1.

Test levels: system test through the HTTP API (preferred for repeatability) with a small set of UI-level checks for the confirmation screen and role-specific menus (TC-01, TC-11, TC-14).

Area	Approach	Test cases
Functional	Scripted request/response checks with state verification in the data store; success and alternate flows A1–A4 of UC-03.	TC-01 ... TC-15, TC-24, TC-25
Non-functional – performance	Fire the A-03 load at the reservation endpoint and record per-request duration; verify against 200 ms.	TC-16
Non-functional – concurrency / integrity	Submit N identical-slot requests simultaneously; verify exactly one confirmation and one stored record.	TC-17
Non-functional – access control, session	See §5.1.	TC-18, TC-19
Security	See §5.1.	TC-11, 12, 18 – 23

5.1 Security Validation

Objective: demonstrate that each security requirement of SRS §4.2 is enforced, and therefore that each security objective of §4.1 is preserved.

Security objective	Security requirement	Control (Architecture §7)	Validation method	Test case	Pass criterion
SEC-OBJ-01 Authenticated & authorized access, SEC-OBJ-02 Integrity	SEC-REQ-01	C-04 Access Control Guard	Call every technician-only endpoint with a Student session	TC-18	HTTP 403 each time; data unchanged
SEC-OBJ-01	SEC-REQ-02	C-03 Authentication & Session Service	Use a session idle for 29 and 31 minutes (test clock)	TC-19	29 min → success; 31 min → HTTP 401
SEC-OBJ-01	SEC-REQ-04	C-02 security filter, C-03	Call every protected endpoint with no token and with a malformed token	TC-20	HTTP 401 for every endpoint; no data returned
SEC-OBJ-01	SEC-REQ-07	C-03 Authentication & Session Service; C-10 / §11.4 logging rules	Compare login error text for unknown user vs. wrong password; inspect logs, responses and stored credentials	TC-11, TC-20	Identical error text; no plaintext credential in log, response or store
SEC-OBJ-03 Confidentiality of own data	SEC-REQ-03	C-04 + C-05 ownership check	Student B reads / cancels Student A's reservation	TC-12, TC-23	HTTP 403; no content leaked; reservation unchanged
SEC-OBJ-02 Integrity	SEC-REQ-05	C-02 input validation, C-11 parameterised access	Boundary, malformed and injection-style inputs	TC-21	HTTP 400/422; no data change; tables intact
SEC-OBJ-04 Accountability	SEC-REQ-06	C-10 Audit Logger	Perform privileged actions, then read audit records	TC-22	One record per action with user, role, action, target, timestamp

Security test notes: tests use only the system under test in the lab environment; no third-party service is attacked. Injection inputs are limited to a short set of standard strings (e.g. ' OR '1'='1, '; DROP TABLE equipment; --).

5.2 Entry, Exit and Suspension Criteria

- **Entry:** build deployed to the test environment; test data loaded (§7); Notification stub and test clock available.
- **Exit:** all test cases executed; all High-priority requirements' tests passed; no open defect that violates a requirement acceptance criterion.
- **Suspension:** environment unavailable, or a blocking defect in login / reservation prevents further testing; resume after fix and re-run of TC-01 and TC-11.

5.3 Pass / Fail Criteria

A test case *passes* when every expected result in its table is observed. It *fails* if any is not. Statuses: *Not Executed*, *Pass*, *Fail*, *Blocked*. Performance (TC-16) passes only if **every** measured request is ≤ 200 ms (the SRS gives no percentile).

6. Test Environment

Item	Specification
Server	Instance of the portal with a relational database supporting transactions and locking (SRS EIR-03); stack not yet chosen.
Client	Any current web browser for UI checks; HTTP client for API checks.
Time	A test clock that can set "now" (needed for TC-03, 07, 08, 09, 19). A-T2
Notification System	Stub that records requests and can be switched to <i>fail</i> .
Network	Single machine or LAN; latency not part of NFR-001 (server-side processing time is measured).

7. Test Data

Data	Content
Users	studentA, studentB, student01...student20 (role Student); tech1 (role Lab Technician). Passwords set by the tester; not recorded in this document.
Equipment	EQ-01 <i>Oscilloscope</i> – Valid; EQ-02 <i>Oscilloscope</i> – Valid (alternative); EQ-03 <i>Logic Analyzer</i> – Expired; EQ-04 <i>FPGA Board</i> – Pending; EQ-05 <i>Logic Analyzer</i> – Valid (only loaded for TC-05 and TC-24).
Reservations	Created by the tests themselves; the store is reset to the baseline before each test case.
Time values	Defined relative to "now" (e.g. <i>now</i> + 1 day 10:00).

8. Test Tools

No tooling has been selected because the stack is not chosen. Required capabilities: an HTTP client able to send scripted requests, a facility to send concurrent requests and record timings (for TC-16, TC-17), direct read access to the data store for verification, and a stub for the Notification System. The tools will be named in the execution report; none are asserted here.

9. Responsibilities

Role	Person
Test designer, executor, reporter	Anagha N (PES1UG24CS058) – individual project
Reviewer / approver	Course instructor

10. Risks, Assumptions and Dependencies

ID	Item
R-1	NFR-001 "peak load" undefined → 20 concurrent requests used (A-03); result may need re-baselining.
R-2	Boundary tests need a controllable clock (A-T2); without it TC-03, 07, 08, 09, 19 cannot be executed precisely.
R-3	Assumptions A-04...A-09 (SRS §2.6) affect expected results; if the instructor decides otherwise, TC-02, 07, 09, 14 must be revised.
R-4	No implementation exists; all schedule and effort depends on it. No dates are promised here.
D-1	Notification stub available (A-12). D-2 Test data reset mechanism.

11. Test Deliverables

This plan; executed test-case table with actual results; defect log; a requirements-to-test traceability report (§13).

12. Test Cases

Fields: Test Case ID, Requirement ID(s), Objective, Preconditions, Test Data, Steps, Expected Result, Actual Result, Pass/Fail (status), Test Type. Times use the test clock; "now" is the clock's current value.

Summary

ID	Requirement(s)	Objective (short)	Type	Status
TC-01	FR-001, FR-002	Successful reservation	Functional	Not Executed
TC-02	FR-001	Duration boundary 120 / 121 min	Functional, Boundary	Not Executed
TC-03	FR-001	7-day horizon boundary	Functional, Boundary	Not Executed
TC-04	FR-001	Overlapping slot on same equipment	Functional, Negative	Not Executed
TC-05	FR-002	Expired / Pending calibration blocked	Functional, Negative	Not Executed
TC-06	FR-003	Cancellation before cut-off	Functional	Not Executed
TC-07	FR-003	Cancellation cut-off boundary	Functional, Boundary	Not Executed
TC-08	FR-004	Late return sets penalty flag	Functional	Not Executed
TC-09	FR-004	On-time / at-end return has no flag	Functional, Boundary	Not Executed
TC-10	FR-005	Inventory add / update / remove	Functional	Not Executed
TC-11	FR-006, SEC-REQ-07	Valid and invalid login	Functional, Security	Not Executed
TC-12	FR-007, SEC-REQ-03	Own history only	Functional, Security	Not Executed
TC-13	FR-008, FR-002	Calibration status update takes effect	Functional	Not Executed
TC-14	FR-009	Confirmation + notification, notification failure	Functional, Fault injection	Not Executed
TC-15	FR-010	Student self-overlap rejected	Functional, Negative	Not Executed
TC-16	NFR-001	Reservation latency ≤ 200 ms	Non-functional (Performance)	Not Executed
TC-17	NFR-003, FR-001	Concurrent requests, one winner	Non-functional (Concurrency)	Not Executed
TC-18	NFR-002, SEC-REQ-01	Student blocked from technician functions	Security	Not Executed

ID	Requirement(s)	Objective (short)	Type	Status
TC-19	NFR-002, SEC-REQ-02, FR-006	30-minute session expiry	Security, Boundary	Not Executed
TC-20	SEC-REQ-04, SEC-REQ-07, FR-006	Unauthenticated access, credential handling	Security	Not Executed
TC-21	SEC-REQ-05, FR-001, FR-005	Invalid input and injection strings	Security, Negative	Not Executed
TC-22	SEC-REQ-06, FR-004, FR-005, FR-008	Audit records for privileged actions	Security	Not Executed
TC-23	SEC-REQ-03, FR-003	Cancel another student's reservation	Security, Negative	Not Executed
TC-24	FR-001	Availability display and filters	Functional	Not Executed
TC-25	FR-005	Remove equipment that has a future reservation	Functional, Negative	Not Executed

Detailed test cases

TC-01 – Successful reservation (Functional)

Field	Content
Requirement ID	FR-001(e), FR-002(a) (UC-03 main flow)
Objective	A student reserves a slot on Valid equipment and receives a token.
Preconditions	studentA logged in; EQ-01 Valid; no reservations exist.
Test Data	EQ-01; start = now + 1 day 10:00; duration 90 min.
Steps	1. Open availability list (UC-02) and select EQ-01. 2. Submit reservation. 3. Open the availability list again.
Expected Result	Reservation created (HTTP 201) with status Confirmed and a non-empty token; confirmation screen shows token and slot; availability shows 10:00–11:30 as reserved; one reservation record stored.
Actual Result	–
Pass/Fail	Not Executed

TC-02 – Duration boundary (Functional, Boundary)

Field	Content
Requirement ID	FR-001(b)
Objective	Slot of exactly 120 minutes is accepted; 121 minutes is rejected.
Preconditions	studentA logged in; EQ-01 Valid, free.
Test Data	start = now + 1 day 10:00; durations 120 and 121 min (separate requests, store reset between).
Steps	1. Reserve with 120 min. 2. Reset. 3. Reserve with 121 min.
Expected Result	120 min → HTTP 201 Confirmed. 121 min → HTTP 422 DURATION_EXCEEDS_LIMIT; no reservation record created.
Actual Result	–
Pass/Fail	Not Executed

TC-03 – Booking horizon boundary (Functional, Boundary)

Field	Content
Requirement ID	FR-001(c)
Objective	A start time exactly 7 days after the request is accepted; later is rejected.
Preconditions	studentA logged in; EQ-01 Valid, free; test clock fixed at T.
Test Data	start = T + 7 days; start = T + 7 days + 1 min; duration 60 min.
Steps	1. Reserve with start = T + 7 d. 2. Reset. 3. Reserve with start = T + 7 d + 1 min.
Expected Result	First → HTTP 201. Second → HTTP 422 <code>START_OUT_OF_RANGE</code> ; no record created.
Actual Result	–
Pass/Fail	Not Executed

TC-04 – Overlapping slot on the same equipment (Functional, Negative; UC-03 alternate flow A2)

Field	Content
Requirement ID	FR-001(d)
Objective	A second student cannot reserve a slot that overlaps a Confirmed reservation of the same equipment.
Preconditions	studentA has Confirmed EQ-01 10:00–11:00 (day D). studentB logged in.
Test Data	studentB requests EQ-01, D 10:30, 60 min. Also D 11:00, 60 min (adjacent).
Steps	1. studentB submits the overlapping request. 2. studentB submits the adjacent request 11:00–12:00.
Expected Result	Step 1 → HTTP 409 <code>SLOT_CONFLICT</code> with next available slots listed; no new record. Step 2 → accepted (adjacent, not overlapping).
Actual Result	–
Pass/Fail	Not Executed

TC-05 – Expired / Pending calibration blocks reservation (Functional, Negative; UC-03 alternate flow A1)

Field	Content
Requirement ID	FR-002(b), FR-002(c)
Objective	Equipment whose calibration is not Valid cannot be reserved; an alternative is suggested.
Preconditions	studentA logged in; EQ-03 Expired; EQ-04 Pending; EQ-05 (<i>Logic Analyzer</i> , Valid) exists as the alternative for the suggestion check.
Test Data	Request for EQ-03 and for EQ-04, tomorrow 10:00, 60 min.
Steps	1. Reserve EQ-03. 2. Reserve EQ-04.
Expected Result	Both → HTTP 409 <code>CALIBRATION_INVALID</code> with message "This equipment is currently unavailable for booking — calibration required."; the EQ-03 response suggests EQ-05 and the EQ-04 response suggests none (no other FPGA board); no reservation record created.
Actual Result	–
Pass/Fail	Not Executed

TC-06 – Cancellation before the cut-off (Functional)

Field	Content
Requirement ID	FR-003
Objective	A student cancels an own reservation; the slot is released within 5 s with no penalty.
Preconditions	studentA has Confirmed EQ-01 reservation starting in 3 hours.
Test Data	The reservation id.
Steps	1. Cancel the reservation. 2. Poll availability of EQ-01 for that slot until Available or 5 s elapse. 3. Check the reservation record.
Expected Result	HTTP 200; status Cancelled; slot shows Available within 5 s; penalty flag false.
Actual Result	–
Pass/Fail	Not Executed

TC-07 – Cancellation cut-off boundary (Functional, Boundary)

Field	Content
Requirement ID	FR-003
Objective	Cancellation exactly 60 min before start is accepted (A-04); at 59 min it is rejected.
Preconditions	Two Confirmed reservations R1, R2 of studentA, both starting at S.
Test Data	Test clock set to S – 60 min for R1; to S – 59 min for R2 (separate runs).
Steps	1. Set clock S – 60 min, cancel R1. 2. Set clock S – 59 min, cancel R2.
Expected Result	R1 → HTTP 200 Cancelled. R2 → HTTP 409 CANCELLATION_WINDOW_CLOSED; R2 remains Confirmed and its slot stays reserved.
Actual Result	–
Pass/Fail	Not Executed

TC-08 – Late return sets the penalty flag (Functional)

Field	Content
Requirement ID	FR-004
Objective	Returning after the slot end records the timestamp and sets the penalty flag.
Preconditions	tech1 logged in; Confirmed reservation with slot end E.
Test Data	Test clock = E + 1 min.
Steps	1. tech1 marks the reservation Returned. 2. Read the reservation.
Expected Result	HTTP 200; status Returned; return timestamp = E + 1 min; penalty flag = true.
Actual Result	–
Pass/Fail	Not Executed

TC-09 – On-time return has no penalty (Functional, Boundary)

Field	Content
Requirement ID	FR-004
Objective	A return before, or exactly at, the slot end is not flagged (A-08).
Preconditions	tech1 logged in; two Confirmed reservations RA, RB with slot end E.
Test Data	Clock = E – 10 min for RA; clock = E for RB.
Steps	1. Return RA at E – 10 min. 2. Return RB at E.
Expected Result	Both: HTTP 200, status Returned, correct timestamps, penalty flag = false.
Actual Result	–
Pass/Fail	Not Executed

TC-10 – Inventory add / update / remove (Functional)

Field	Content
Requirement ID	FR-005
Objective	Technician changes to the inventory are persisted and visible immediately after saving.
Preconditions	tech1 logged in; studentA has a second session open.
Test Data	New item: name "Spectrum Analyzer", category "RF", calibration due date = a future date. Update: category → "Instruments".
Steps	1. Start the add form but do not save; studentA lists equipment. 2. Save the new item; list again. 3. Update the item; list again. 4. Remove the item; list again.
Expected Result	Step 1: item not visible. Step 2: HTTP 201, item listed. Step 3: HTTP 200, new category listed. Step 4: HTTP 200, item no longer listed. Changes persist after re-reading from the store.
Actual Result	–
Pass/Fail	Not Executed

TC-11 – Valid and invalid login (Functional, Security)

Field	Content
Requirement ID	FR-006, SEC-REQ-07
Objective	Valid credentials give a session with the correct role; invalid credentials give no session and a generic error.
Preconditions	Accounts studentA, tech1 exist.
Test Data	Valid pairs for both users; unknown user nobody /any password; studentA with a wrong password.
Steps	1. Log in as studentA. 2. Log in as tech1. 3. Log in as nobody. 4. Log in as studentA with wrong password.
Expected Result	1–2: HTTP 200, session issued, role Student / Lab Technician, role-appropriate menu. 3–4: HTTP 401 INVALID_CREDENTIALS, identical message and no session in both cases.
Actual Result	–
Pass/Fail	Not Executed

TC-12 – Own reservation history only (Functional, Security)

Field	Content
Requirement ID	FR-007, SEC-REQ-03
Objective	A student sees all and only their own reservations with the required fields.
Preconditions	studentA has 2 reservations (one Confirmed, one Cancelled); studentB has 1.
Test Data	–
Steps	1. studentA opens "My reservations". 2. studentA requests the detail of studentB's reservation id directly.
Expected Result	1: exactly 2 entries, each with equipment, start/end, status, token, penalty flag. 2: HTTP 403 FORBIDDEN_RESOURCE; no data of studentB's reservation in the response.
Actual Result	–
Pass/Fail	Not Executed

TC-13 – Calibration status update takes effect (Functional)

Field	Content
Requirement ID	FR-008, FR-002
Objective	A status change by the technician governs subsequent reservation requests.
Preconditions	tech1 and studentA logged in; EQ-01 Valid.
Test Data	EQ-01 → Expired → Valid; request tomorrow 10:00, 60 min.
Steps	1. tech1 sets EQ-01 Expired. 2. studentA reserves EQ-01. 3. tech1 sets EQ-01 Valid. 4. studentA reserves EQ-01 again. 5. tech1 sends status "Unknown".
Expected Result	1: HTTP 200. 2: HTTP 409 CALIBRATION_INVALID. 3: HTTP 200. 4: HTTP 201 Confirmed. 5: HTTP 422 INVALID_CALIBRATION_STATUS, status unchanged.
Actual Result	–
Pass/Fail	Not Executed

TC-14 – Confirmation and notification, incl. notification failure (Functional, Fault injection)

Field	Content
Requirement ID	FR-009 (A-06)
Objective	A confirmed reservation produces an on-screen confirmation and one notification request; notification failure does not remove the reservation.
Preconditions	studentA logged in; EQ-01 Valid; Notification stub recording.
Test Data	Two reservations on different free slots; stub in <i>success</i> mode for the first, <i>fail</i> mode for the second. Also one rejected request (calibration invalid).
Steps	1. Reserve (stub OK). 2. Reserve a different slot (stub fails). 3. Attempt a rejected reservation (EQ-03).
Expected Result	1: confirmation shown; stub received exactly one request with student, equipment, slot, token. 2: confirmation still shown, reservation Confirmed, failure entry in the application log. 3: stub received no request.
Actual Result	–
Pass/Fail	Not Executed

TC-15 – Student self-overlap rejected (Functional, Negative)

Field	Content
Requirement ID	FR-010
Objective	One student cannot hold two overlapping reservations on different equipment.
Preconditions	studentA has Confirmed EQ-01 on day D 10:00–11:00; EQ-02 Valid, free.
Test Data	studentA requests EQ-02, D 10:30, 30 min.
Steps	1. Submit the request.
Expected Result	Rejected with HTTP 409 STUDENT_SLOT_CONFLICT; no new record; EQ-02 slot remains available.
Actual Result	–
Pass/Fail	Not Executed

TC-16 – Reservation latency (Non-functional – Performance)

Field	Content
Requirement ID	NFR-001
Objective	Reservation requests are processed in ≤ 200 ms under the A-03 test load.
Preconditions	Baseline data; accounts student01 ... student20 logged in; timing facility ready.
Test Data	20 concurrent requests, one per student, each for a different non-overlapping slot of EQ-01 or EQ-02 (so every request is legitimate and can succeed).
Steps	1. Send the 20 requests simultaneously. 2. Record the processing time of each (request received → response sent). 3. Repeat the run 5 times.
Expected Result	Every recorded processing time ≤ 200 ms in all runs; all 20 requests per run succeed.
Actual Result	–
Pass/Fail	Not Executed

TC-17 – Concurrent requests for the same slot (Non-functional – Concurrency / data integrity)

Field	Content
Requirement ID	NFR-003, FR-001
Objective	With simultaneous identical-slot requests, exactly one reservation is confirmed.
Preconditions	EQ-01 Valid and free at slot X; accounts student01 ... student20 logged in.
Test Data	20 simultaneous requests for EQ-01, slot X (tomorrow 10:00, 60 min).
Steps	1. Release all 20 requests at the same instant. 2. Count responses by status. 3. Query the store for reservations of EQ-01 at X.
Expected Result	Exactly 1 × HTTP 201; 19 × HTTP 409 SLOT_CONFLICT; exactly 1 stored reservation for the slot; exactly one token issued.
Actual Result	–
Pass/Fail	Not Executed

TC-18 – Student blocked from technician-only functions (Security)

Field	Content
Requirement ID	NFR-002, SEC-REQ-01
Objective	Technician-only endpoints reject a Student session without changing data.
Preconditions	studentA logged in; a Confirmed reservation R and equipment EQ-01 exist.
Test Data	Calls: update calibration of EQ-01 → Expired; add equipment; update equipment; remove equipment; return R.
Steps	1. Send each call with studentA's token. 2. Compare store state before and after.
Expected Result	Each call → HTTP 403 FORBIDDEN_ROLE; EQ-01 status, inventory and R unchanged. Same calls with tech1's token succeed (control).
Actual Result	–
Pass/Fail	Not Executed

TC-19 – Session expiry after 30 minutes of inactivity (Security, Boundary)

Field	Content
Requirement ID	NFR-002, SEC-REQ-02, FR-006
Objective	A session idle 29 min still works (and its idle timer restarts); one idle 31 min is rejected.
Preconditions	Test clock available; two logins of studentA (sessions S1, S2).
Test Data	S1: idle 29 min then request, then idle 29 min again then request. S2: idle 31 min then request.
Steps	1. Advance clock 29 min; S1 requests availability. 2. Advance 29 min; S1 requests again. 3. Advance S2's idle time to 31 min; S2 requests availability. 4. Log in again.
Expected Result	1 and 2: HTTP 200 (activity resets the timer). 3: HTTP 401 SESSION_EXPIRED, no data returned. 4: new session works.
Actual Result	–
Pass/Fail	Not Executed

TC-20 – Unauthenticated access and credential handling (Security)

Field	Content
Requirement ID	SEC-REQ-04, SEC-REQ-07, FR-006
Objective	Protected functions need a valid session; credentials are not exposed.
Preconditions	None logged in.
Test Data	Every protected endpoint of Design 99 called with (a) no token, (b) a malformed token. Login attempts with a known password.
Steps	1. Call each protected endpoint without a token, then with a malformed token. 2. Perform successful and failed logins. 3. Inspect the application log, audit records and the stored user record.
Expected Result	1: HTTP 401 AUTHENTICATION_REQUIRED for every endpoint, no data (SEC-REQ-04). 3: the password does not appear in any log or response, and is not stored as plain text (SEC-REQ-07).
Actual Result	–
Pass/Fail	Not Executed

TC-21 – Invalid input and injection strings (Security, Negative)

Field	Content
Requirement ID	SEC-REQ-05, FR-001, FR-005
Objective	Malformed or hostile input is rejected on the server and does not change data.
Preconditions	studentA and tech1 logged in; baseline data.
Test Data	Reservation: duration 0, -30, "abc"; start in the past; missing equipmentId; equipmentId 1 OR 1=1. Availability filter category=' OR '1'='1. Inventory: empty name; name = '; DROP TABLE equipment; - .
Steps	1. Send each request. 2. After all requests, read equipment and reservation tables and the equipment list.
Expected Result	Reservation/filter/validation cases → HTTP 400/422 with a field-level error and no state change; the injection filter returns no extra rows. The inventory name containing SQL text is either rejected by validation or stored literally as text. All tables and baseline rows remain intact.
Actual Result	–
Pass/Fail	Not Executed

TC-22 – Audit records for privileged actions (Security)

Field	Content
Requirement ID	SEC-REQ-06, FR-004, FR-005, FR-008
Objective	Each privileged action creates an attributable audit record.
Preconditions	tech1 logged in; empty audit log; Confirmed reservation R.
Test Data	Actions: set EQ-01 calibration Expired; add equipment; update equipment; remove equipment; return R.
Steps	1. Perform the five actions. 2. Read the audit records.
Expected Result	Exactly 5 records, each with user id (tech1), role, action type, target id, timestamp (≈ action time). Failed/rejected attempts (TC-18) create no record of success.
Actual Result	–
Pass/Fail	Not Executed

TC-23 – Cancel another student's reservation (Security, Negative)

Field	Content
Requirement ID	SEC-REQ-03, FR-003
Objective	A student cannot cancel a reservation owned by someone else.
Preconditions	studentA owns Confirmed reservation R (starts in 3 hours); studentB logged in.
Test Data	R's id.
Steps	1. studentB sends the cancel request for R. 2. Read R.
Expected Result	HTTP 403 FORBIDDEN_RESOURCE ; R remains Confirmed and its slot stays reserved.
Actual Result	–
Pass/Fail	Not Executed

TC-24 – Availability display and filters (Functional)

Field	Content
Requirement ID	FR-001(a)
Objective	The availability list shows equipment with its calibration status, the filters work, and reserved slots equal the Confirmed reservations.
Preconditions	Baseline equipment EQ-01...EQ-05 (\$7); studentA logged in; EQ-01 has one Confirmed reservation R1 (day D 10:00–11:00) and one Cancelled reservation R2 (day D 14:00–15:00).
Test Data	Filters: category=Oscilloscope; calibrationStatus=Expired; both together; availability of EQ-01 for day D.
Steps	1. List equipment without filter. 2. List with category=Oscilloscope. 3. List with calibrationStatus=Expired. 4. List with both filters. 5. Request availability of EQ-01 for day D.
Expected Result	1: five items, each with its calibration status. 2: EQ-01 and EQ-02 only. 3: EQ-03 only. 4: empty list (empty-list message in the UI). 5: exactly one reserved slot, 10:00–11:00 (R2 is not shown as reserved).
Actual Result	–
Pass/Fail	Not Executed

TC-25 – Remove equipment that has a future reservation (Functional, Negative; depends on assumption A-07)

Field	Content
Requirement ID	FR-005
Objective	Equipment with a future Confirmed reservation cannot be removed; equipment without one can.
Preconditions	tech1 logged in; EQ-01 has a future Confirmed reservation R; EQ-02 has none.
Test Data	Remove EQ-01; remove EQ-02.
Steps	1. Remove EQ-01. 2. Read EQ-01 and R. 3. Remove EQ-02. 4. List equipment.
Expected Result	1: HTTP 409 EQUIPMENT_HAS_ACTIVE_RESERVATIONS. 2: EQ-01 still listed, R still Confirmed. 3: HTTP 200. 4: EQ-02 no longer listed. If the instructor decides otherwise for A-07, this case is revised.
Actual Result	–
Pass/Fail	Not Executed

13. Traceability Matrix (Requirement → Test Case)

Requirement	Test cases	Coverage note
FR-001	TC-01, TC-02, TC-03, TC-04, TC-17, TC-21, TC-24	clause (a) TC-24; (b) TC-02; (c) TC-03; (d) TC-04, TC-17; (e) TC-01; invalid input TC-21
FR-002	TC-01, TC-05, TC-13	clause (a) TC-01, TC-13; (b), (c) TC-05
FR-003	TC-06, TC-07, TC-23	
FR-004	TC-08, TC-09, TC-22	
FR-005	TC-10, TC-21, TC-22, TC-25	
FR-006	TC-11, TC-19, TC-20	
FR-007	TC-12	
FR-008	TC-13, TC-22	
FR-009	TC-14	
FR-010	TC-15	

Requirement	Test cases	Coverage note
NFR-001	TC-16	
NFR-002	TC-18, TC-19	
NFR-003	TC-17	
SEC-REQ-01	TC-18	
SEC-REQ-02	TC-19	
SEC-REQ-03	TC-12, TC-23	
SEC-REQ-04	TC-20	
SEC-REQ-07	TC-11, TC-20	
SEC-REQ-05	TC-21	
SEC-REQ-06	TC-22	

Totals: 25 test cases – 17 functional (TC-01...TC-15, TC-24, TC-25), 2 non-functional (TC-16, TC-17), 6 security (TC-18...TC-23); TC-11 and TC-12 also carry security requirements. Every requirement in the SRS has at least one test case. Test types present: normal-flow, invalid-input, boundary, error-handling, fault-injection, performance, concurrency and security.